

A Multi-Method Machine Learning Evaluation of Mortality in the United States

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Introduction

Predicting all cause mortality is a fundamental human goal with wide-ranging implications from medicine to public health to the life insurance industry. Within the life insurance industry, risk assessment of mortality at the individual level can guide pricing decisions, while understanding population level trends can guide macro-level financial planning. The clinical literature on all cause mortality is vast. Pubmed alone has indexed over 14,000 clinical studies tagged with this term. Synthesizing this large and growing web of knowledge into a tool that can be deployed for pricing presents a barrier to efficient life insurance markets, and a competitive advantage to firms which can overcome such a challenge. Machine learning offers promise as a potential solution to this problem given its ability to handle large datasets, find both linear and nonlinear relationships, and extract patterns from both labelled and unlabelled data.

This project uses a variety of machine learning methods to perform a broad sweep of analyses on publicly available mortality data for the United States population. Supervised methods were used on both individually labelled mortality data, and aggregated population level mortality rates for classification and regression tasks, respectively. Unsupervised learning was performed on the aggregate population level data search for useful cluster archetypes. The combination of supervised and supervised methods with both individual and population data goes beyond the scope of prior published works which have focused on one family of models such as classifiers or regressors only. Among all the methods used, there was consensus that age is the predominant predictor of mortality, with gender and health indicators having comparatively modest impact.

Related work

The existing literature provides several examples of similar projects which informed the design of this study. Levantesi & Pizzorusso (2019) used decision trees, random forest, and gradient boosted trees to predict the overall mortality rate among the Italian population using data from the Human Mortality Database [1]. Feature selection was based on available demographics such as age and gender. While our study also performs regression on population level mortality, we add additional features by using the NHANES dataset as a proxy for population level health markers. Qui et al (2022) used gradient boosted trees to predict 5-year mortality among NHANES subjects [2]. Features were selected by hierarchical supervised distance clustering and retaining the most important features in each cluster. Our study differs by using a 10-year endpoint for classification tasks, and by including regression tasks. Our study also differs by using unsupervised clustering tasks. Williams et al (2020) used logistic regression, decision tree, and random forest classifiers to predict mortality among NHANES subjects [3]. Feature selection was done using established risk factors for all-cause mortality such as smoking and diabetes. Our study differs by using additional regression methods and incorporating unsupervised learning. Together, these studies provide an overview of the types of publicly available mortality datasets that exist and a framework for matching datasets to appropriate machine learning tasks.

Data Sources

The **National Health and Nutrition Examination Survey (NHANES)** recurs on a biennial cycle and collects many features from physical examinations, clinical laboratory tests, and health behavior interviews. The dataset is located at <https://wwwn.cdc.gov/nchs/nhanes/> and provides data tables as XPT files, with numerous tables supplied for each cycle of the study. Important variables include age, gender, body measurements, laboratory values, and health questionnaire responses such as smoking and drinking history. For this project, 29,986 records were retrieved covering the years 1999–2017. The **CDC Wide-ranging Online Data for Epidemiologic Research (CDC WONDER)**, located at <https://wonder.cdc.gov/>, provides mortality data. Data are also downloadable via an API, with an example implementation available at [this github repository](#). Data retrieved through the API are supplied in XML format, and important features include year, age, gender, number of deaths, and population. A total of 4,488 records were retrieved covering 1999–2020. The **National Death Index (NDI)** is located at <https://www.cdc.gov/nchs/ndi/index.html>, with one data file for each NHANES cycle supplied as a fixed-width .DAT file. Important variables include mortality outcome (boolean) and time from the NHANES interview measured in months. For this project, 101,316 records were retrieved covering years 1999–2017.

Preprocessing & Feature engineering

The NHANES dataset was the largest and required the most preprocessing. For each cycle of the study, features are spread across multiple files for a major topic. For instance, all demographic data for the 1999-2000 cycle are in the 1999-2000 DEMO data file. A combined dataset was created by downloading each included data table name and year of the NHANES study. The choice of tables to include was informed by prior studies which used demographic variables (age, race, gender), clinical variables which contribute to metabolic syndrome such as (weight, BMI, blood glucose, cholesterol), and smoking history. Tables were merged on “SEQN” subject identifier then cycles of the study were concatenated vertically. Some features names, such as HDL cholesterol, had varying names across iterations of the study, which were consolidated. Others, such as blood pressure, were duplicated with “_number” suffix to indicate serial measurements. From these, an averaged feature with _mean suffix was created. Categorical variables for ethnicity, education level, and self-reported smoking history were converted to dummy variables. Features with >60% missing values were dropped. For population-level analyses data were grouped by year, age, and gender to match WONDER and aggregated by mean, except “SEQN” which was aggregated to subject count. The CDC-WONDER and NDI were separately merged onto NHANES dataset largely unchanged. Some features such as gender in the CDC-WONDER data needed to be mapped to match encodings used in NHANES. NDI and NHANES were merged on “SEQN”. The final NHANES+WONDER dataset had 1,372 records while the NHANES+NDI dataset had 26,453 records. See [Appendix A](#) complete feature lists. Additional task-specific feature engineering will be discussed within relevant sections.

Supervised Learning

Classification

Methods

Classification was done using the NHANES+NDI merged dataset. To account for variable time to outcome measurement in the NDI mortality labels, a 10-year survival boolean was engineered. This endpoint was chosen to balance the trade-off between increasing class balance and decreasing sample size as survival endpoints get longer. Several features were dropped to prevent target leakage. These include NHANES study year, time to follow up, mortality statistic, and subject id. All remaining features were retained. Missing values were filled with means where required by model. Missing x values were imputed with column means where required by model. A total of 5 classifiers were trained. Each model was first trained with default hyperparameters, then tuned using GridCVsearch.

The first three classifiers were tree based: decision tree (DT), random forest (RF), and gradient boosted tree (GBT). The simplest of these is DT, which is akin to a rule-based flowchart which produces a deterministic result. This makes DT based models desirable when nonlinear relationships are expected, but can be prone to overfitting. RF is an ensemble method which trains many trees on bootstrapped samples to “vote” on a classification result. This makes RF less noisy and less prone to overfitting. GBT is an ensemble which trains trees in sequence, rather than parallel, with each tree trained on the error from the last using gradient descent. This offers more parameters to tune and potentially higher top-end performance. A key parameter for trees is max depth, which caps the longest path of nodes in a tree. The ensemble methods can be tuned by the number of estimators used. GBTs can be further tuned with a learning rate parameter, which controls how strongly the model corrects for loss in each iteration.

Support vector (SVC) and k-nearest neighbor (KNN) classifiers were also trained. SVC seeks to find a decision boundary which separates class clusters with few edge cases. It does this by maximizing the distance from the decision boundary to the nearest edge points. A key hyperparameter for SVC is C, which controls the regularization strength. KNN works by storing the whole dataset and predicting new labels using labels by taking a “vote” from the closest points in the training dataspace. Key parameters for KNN include the number of neighbors assessed and the distance method used.

Evaluation

The tuned models were evaluated with cross validation for precision, recall, f1 score, and ROC AUC. ROC AUC score was used as the primary metric given class imbalance toward the positive case (survival), and desire to minimize both false positives and false negatives. GBT was the strongest model with a slight edge over RF and SVC (Figure 1). GBT also had the highest F1 score of the models (0.92 vs 0.91 and 0.86 for RF and SVC, respective). DT and KNN were fastest to fit but performed worse on evaluation metrics, which presented a tradeoff in speed vs model quality.

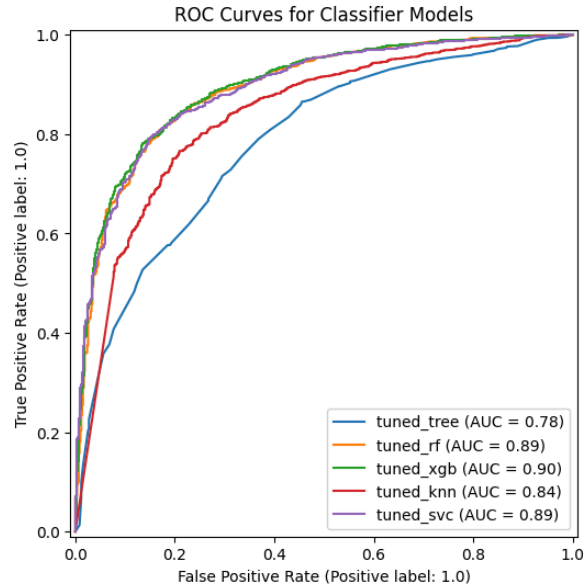


Figure 1. ROC Curves for classifiers.

In depth evaluation of the GBT model revealed age to be the most important feature in the dataset by a wide margin. SHapley Additive ExPlanations (SHAP) analysis showed a gradual decrease in survival with increasing age, while ablation analysis showed a 0.09 point decrease in AUC ROC score when ages are permuted in the training set (Figure 2). Sensitivity analysis was done using a learning curve, which plateaus sharply at training set size of about 1000 (Figure 3). This indicates the training data was sufficiently large for optimal performance.

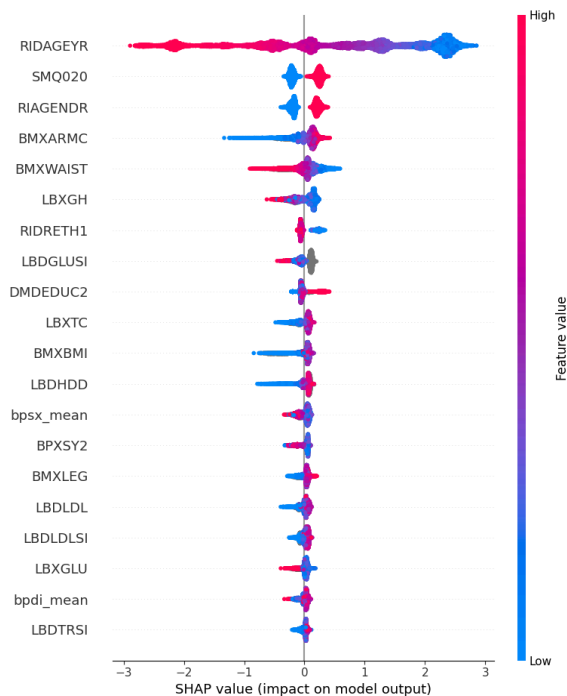


Figure 2. SHAP analysis of GBT classifier.

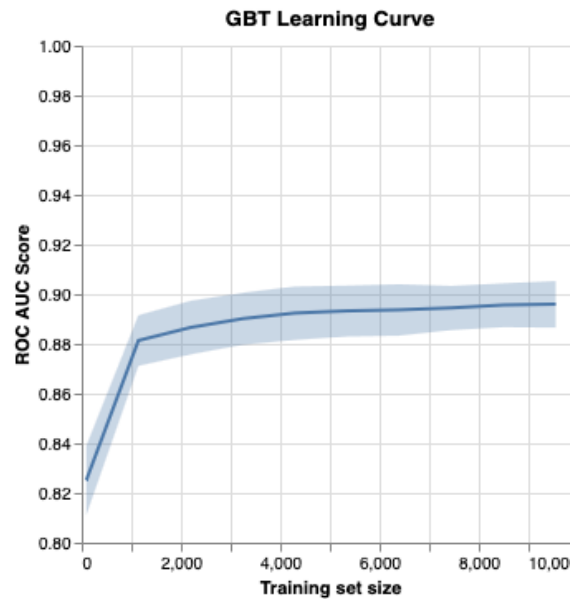


Figure 3. GBT learning curve with standard deviation banding

Failure analysis was done by reviewing three randomly sampled cases where the model predicted 10-year survival incorrectly:

1. False negative: 55 yo Mexican-American male with less than 9th grade education, BMI 28.6, high blood pressure, high blood sugar, elevated triglycerides and LDL cholesterol
2. False positive: An 81 yo non-hispanic white female with high school/GED level education, normal BMI, high blood pressure, normal lab values
3. False negative 56 yo other/multiracial male with 9-11th grade education, with normal BMI, mildly high blood pressure, and high triglycerides (other labs normal)

Based on these cases, it appears the XGB classifier may not weigh age heavily enough relative to other demographics and lab values. The model may be overly biased toward shorter life expectancy for non-white males. To address false positives among those with advanced age, additional feature engineering may help capture interactions between age and other factors. To address false negatives among non-white subjects, weighting could be used to ensure more diverse samples are included in training data. If the data are insufficient, the model could also be trained without ethnicity as a training feature.

Regression

4 different regression models tested and considered:

1. **Poisson Regression** - Used to predict data count where variance is equal to the mean
2. **Negative Binomial Regression** - Used to predict count data where variance is greater than the mean (overdispersion).
3. **Random Forest Regressor** - Tree-based ensemble. Used for complex, non-linear relationships
4. **K-Nearest Neighbors (KNN) Regression** - Instance and. Distance-based. Used for capturing simpler non-linear data

Feature Selection and Data Preprocessing

To compare age effects across years fairly, data was filtered on age 20-80 to compare age effects across years fairly due to data limitation - age 85 was only available from 1999-2005 and age 12-19 was only available on 2005-2017 on NHANES dataset. We expect the impact of dropping <20 and 80+ to be immaterial and won't impact the conclusion of our analysis. We've selected features that we think will have the most impact on mortality such as BMI, Age, Gender, Year, Smoker Status, and Race and we filtered out the some features to ensure there is no multicollinearity in the dataset by calculating Variance Inflation Factor (VIF).

Evaluation Metrics

We've used Root Mean Square Error (RSME) "measures average error penalizing large mistakes heavily", Mean Absolute Error (MAE) "measure average error evenly", and R2 "measures percentage of variance explained by the model compared to a simple average baseline" to evaluate model performance to pick the best model. For the two GLM models (Poisson and Negative Binomial), AIC (Akaike Information Criterion) was used to evaluate the

model fit on the given dataset. For RMSE and MAE, a lower score is better and for R2 and AIC, a higher score is better.

	RMSE*	MAE*	R2*	AIC**
Random Forest	117.823 ± 20.137	51.818 ± 4.846	0.993 ± 0.002	N/A
KNN	481.353 ± 84.652	239.578 ± 30.308	0.878 ± 0.038	N/A
Negative Binomial	246.741 ± 31.632	112.621 ± 13.567	0.968 ± 0.006	17426.938
Poisson	114.259 ± 18.999	61.996 ± 5.419	0.993 ± 0.002	19377.686

**We've used 5 cross-validation folds to calculate the mean and standard deviation.*

***AIC is the final fitted model and not calculated based on the 5-fold CV*

Based on results, **Random Forest was the best model** - it outperformed other models across all evaluation metrics with lowest RMSE & MAE and highest R2 scores. Random forest produces higher and better predictions than traditional regression models because they capture nonlinear relationships between the predictors.

Hyperparameter Tuning

We've performed hyperparameter tuning for the Random Forest and KNN regression models to find the best model parameters. Five-fold cross-validation was used to identify the parameter combination that minimized RMSE.

Before hyperparameter turning, we performed **sensitivity analysis** on the maximum tree depth (3, 5, 10, 20, unlimited) and the RSME values were 120.3 for 10, 120.4 for 20, and 121.2 for unlimited. Due to the immaterial difference between tree depth amongst 10,20 and unlimited, we've decided to cap the maximum tree depth to 20 in hyperparameter tuning considering **identify tradeoffs**. Deeper trees (Max Dept=None) fit training data better, but excessive depth may create overfitting and using more trees (500) increases model stability and performance with better accuracy, but it takes longer training and prediction time.

For the random forest model, number of trees (100, 300, 500), maximum tree depth (5, 10, 20), and minimum leaf size (1, 3, 5) were evaluated. Based on the hyperparameter tuning, maximum tree depth of 20, minimum sample leaf of 1, and number of tree estimators of 500 were chosen as the best parameters.

Hyperparameter tuning for KNN focused on the number of neighbors, weighting scheme, and distance metric. The number of neighbors ranging from 3 to 20, uniform and distance-weighted scheme are considered and evaluated, along with Manhattan and Euclidean distance metrics. Based on the hyperparameter tuning, number of neighbors of 15, distance-weighted scheme, and Manhattan distance metrics were chosen as the best parameters.

Feature Analysis

We've performed an Ablation analysis and Feature Importance analysis on the Random

Forest Model to identify key features that had the highest impact on predicting mortality count. **Ablation analysis (Figure 4)** removes one feature at a time to understand how much the feature contributes to the model performance. Based on the test, age had the highest predictive power for predicting mortality while gender and year were 2nd and 3rd from the list. The **Feature Importance analysis (Figure 5)** validated the results from the Ablation analysis where age was the dominant key feature for predicting mortality while gender and year were 2nd and 3rd from the list.

We've also considered using "clusters" from the unsupervised learning to determine whether the model prediction can be improved. The model performance measures (RSME and MAE) improved slightly but worsened other metrics like overdispersion, AIC, and introduced higher multicollinearity where VIF increased. Therefore, we've decided not to include them when evaluating pros and cons.

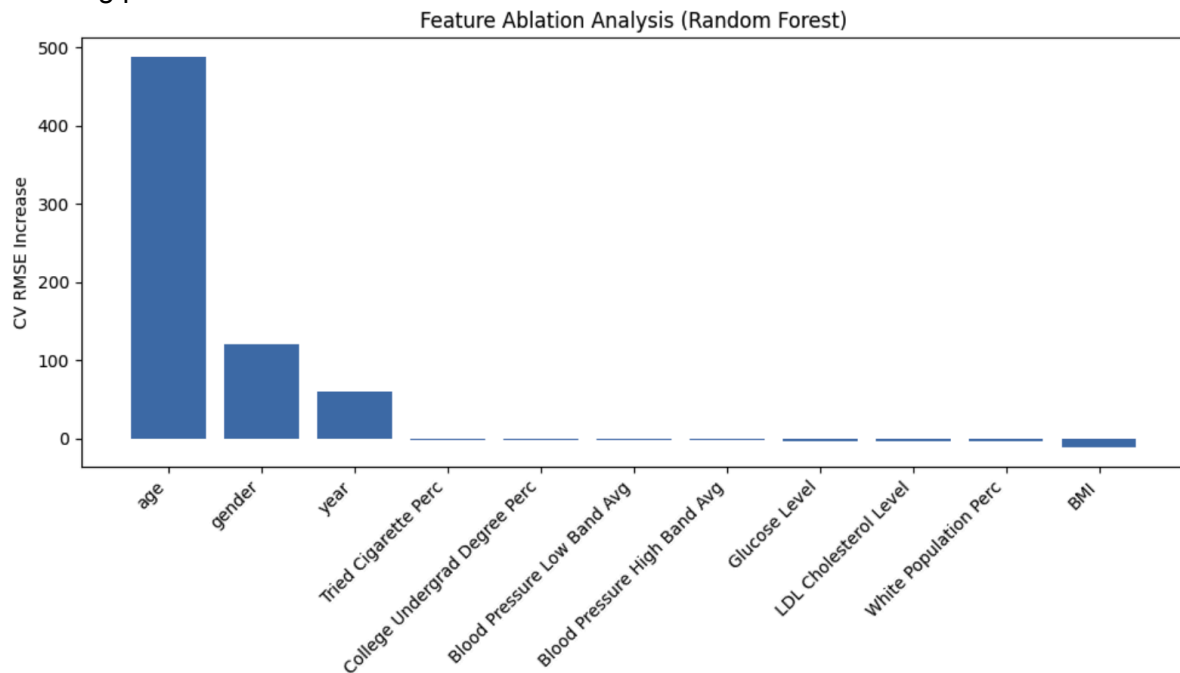


Figure 4. Feature Ablation Analysis for Random Forest Model

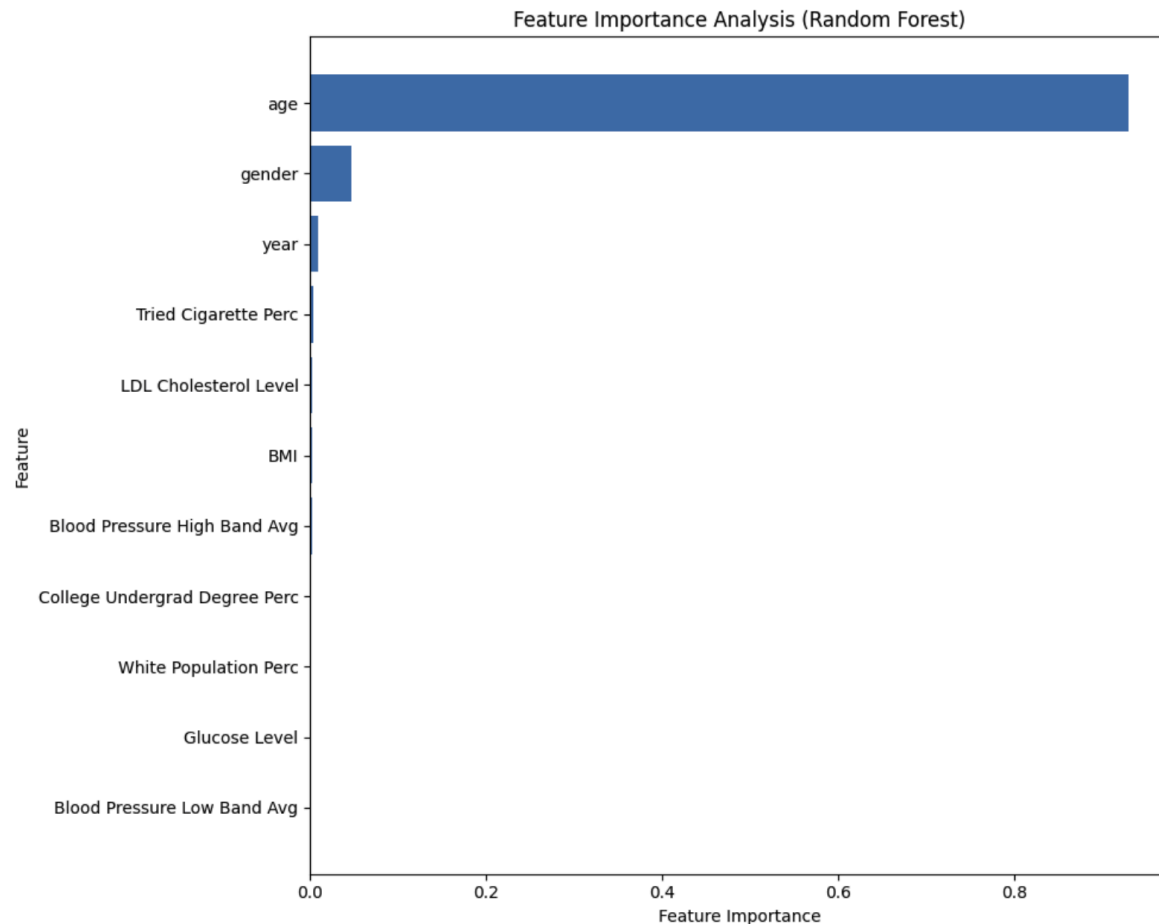


Figure 5. Feature Importance Analysis for Random Forest Model.

Failure Analysis

Comparing Actual and Predicted of the Random Forest model, the highest errors were found in the older age groups. Top 10 highest actual to prediction variance found in age 76-80 group. We expect that this might be caused by greater variance in mortality for older age groups caused by individual factors such as chronic diseases and access to healthcare. These variables were not included in the dataset and the model may not have captured the variability in mortality prediction in older individuals. In addition, the older age population is lower compared to younger or middle age groups due to a smaller surviving population, reducing the amount of data available for the model to learn reliable patterns. Finally, mortality at older ages may be more sensitive to external factors such as seasonal influenza, heat waves, emerging diseases, or changes in healthcare practices, which are not captured by the predictors used in this study. These factors likely contributed to the increased prediction error in the oldest age groups. This error can be classified as a Edge Cases category and to resolve this, we could have spent more time on feature engineering to capture this variance.

A very high overdispersion of 8.85 was detected in the Poisson regression model. To resolve this, we've decided to consider Negative Binomial Regression to accommodate for the wider fluctuations observed. Based on the model performance metrics, the poisson regression model performed better in the evaluation metric (RMSE, MAE, and R2) and AIC, which measures the model fit of the observed data, but it was not a good fit due to a high

overdispersion. This can be classified as a systematic error, where it's driven by a flaw in the model selection and we've resolved it by adding Negative Binomial regression as an alternative.

In the ablation analysis to measure feature relevance of the Random Forest model, we saw a slight negative RMSE increase for certain features like (LDL Cholesterol Level, College Undergraduate Degree Percentage, Tried Cigarette Percentage, and Gender), which means removing these features will improve the model performance. This suggests that the model might be using too many predictors, and some features may be noise and redundant with correlated features. This can be classified as a systematic error, and to resolve this, we could have done more feature engineering, which could have improved the model performance.

Unsupervised Learning

Method Description

In this project, we used unsupervised learning to identify latent population health and mortality profiles, including potential outlier subgroups, across individuals characterized by health-related and demographic factors. The resulting cluster assignments may also serve as features in a subsequent supervised learning phase for downstream prediction tasks. The workflow consisted of four stages:

1. **Feature Selection and Preprocessing:** Highly correlated variables were identified using a correlation threshold of 0.85 and removed to reduce redundancy, resulting in 13 retained health-related features. Age and sex were kept as key demographic variables given their well-established associations with mortality risk and their relevance for forming clinically meaningful subgroups. Year was excluded to prevent temporal effects from influencing cluster formation. Other demographic variables — including race/ethnicity, educational attainment, and smoking history — along with mortality-related outcome measures were excluded from the clustering model to avoid confounding and outcome leakage, and were instead reserved for post-hoc profiling and interpretation. All retained features were standardized using z-score normalization to ensure comparability across variables in distance-based clustering algorithms.
2. **Dimensionality Reduction and Feature Representation:** Given the relatively low dimensionality of the retained feature set and the prior removal of highly correlated variables, dimensionality reduction was not applied during clustering. The first two principal components explained 59.1% of total variance, which is sufficient for visualization but not enough to replace the full feature space without losing information. Preserving the original feature space also allowed for more direct interpretation of cluster characteristics and feature contributions. PCA was therefore used only for visualization, projecting the data into two dimensions to facilitate visual assessment of cluster structure and separation.
3. **Clustering Analysis:** Three unsupervised learning methods with different underlying assumptions were applied to capture different aspects of the structure in the data. K-Means was selected as a centroid-based method that assumes roughly spherical clusters and provides a useful baseline for partitioning observations into distinct health

profiles. Hierarchical Agglomerative Clustering was used to explore potential nested or multi-scale structure without requiring a predefined number of clusters, making it possible to examine cluster stability across resolutions. DBSCAN was included as a density-based method capable of identifying irregularly shaped clusters and detecting outliers, which is particularly relevant given that some individuals may have unusual or extreme health profiles.

4. **Hyperparameter Tuning and Model Selection:** Tuning was conducted separately for each method. For K-Means, multiple values of k were evaluated using the Elbow Method to assess within-cluster variance. For hierarchical clustering, the dendrogram was examined to identify natural cut points at different levels of granularity. For both methods, the Silhouette Score and Davies–Bouldin Index (DBI) were also used to compare cluster quality in terms of cohesion and separation. For DBSCAN, combinations of ϵ and min_samples were explored using k -distance plots, with final parameters selected based on cluster quality, interpretability, and the meaningfulness of detected outliers.

Unsupervised Evaluation

We first applied K-Means and evaluated the optimal number of clusters using the Elbow Method, Silhouette Score, and Davies–Bouldin Index (DBI). The Silhouette Score measures cluster separation (higher is better), while the DBI evaluates compactness and separation together (lower is better).

As shown in Figure 6, the three metrics did not converge on a single value of k . The Silhouette Score and DBI both favored $k = 3$, while the Elbow Method indicated a possible inflection around $k = 4$. This disagreement suggests the data may have both broad group-level structure and finer substructure within those groups — a smaller k captures the overall separation, while a slightly larger k may reflect within-group variation.

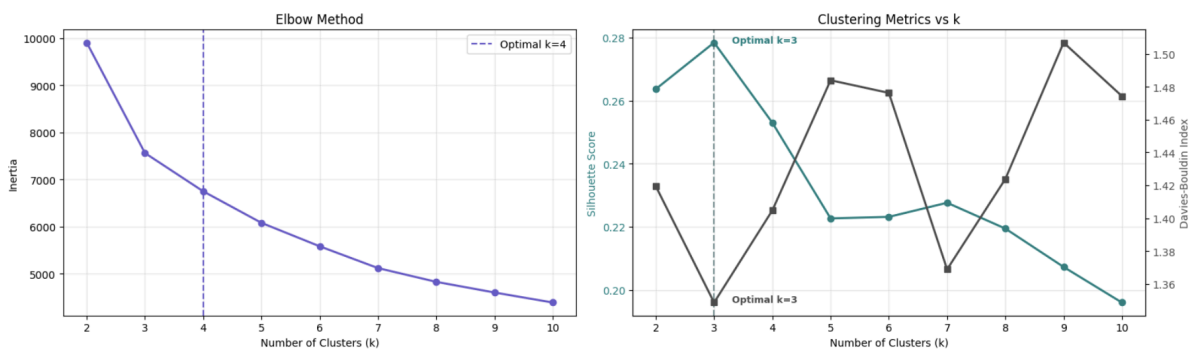


Figure 6. Elbow method, Silhouette Score, and Davies–Bouldin Index (DBI) used to determine the optimal number of clusters (K) for K-Means clustering.

We then applied hierarchical clustering, evaluated using a dendrogram, Silhouette Score, and DBI. Unlike K-Means, all three approaches pointed to the same solution. The Silhouette Score peaked at $k = 3$, the DBI was minimized at $k = 3$, and the dendrogram showed three well-separated branches with a large distance gap between the three- and four-cluster solutions, indicating a stable cut at $k = 3$. Together, these results provide stronger support for

a three-cluster partition.

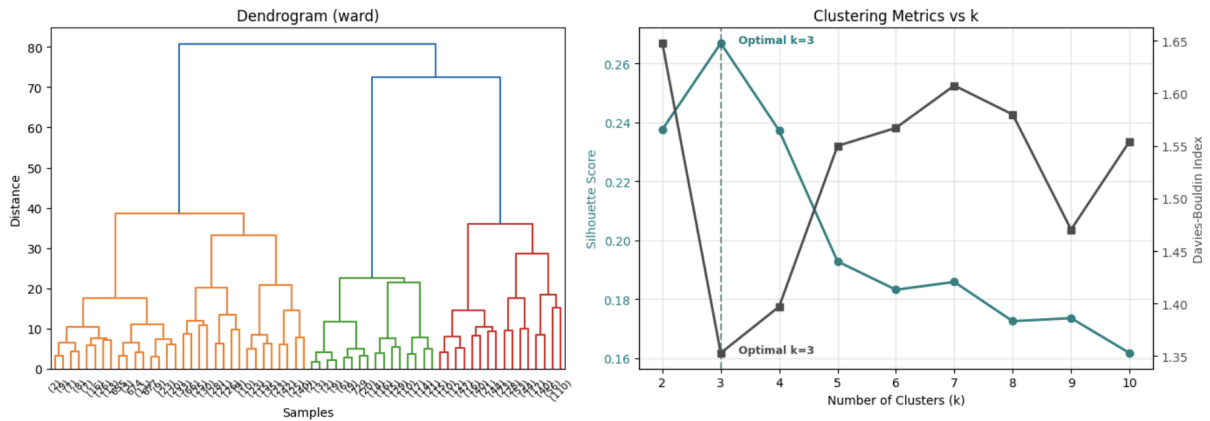


Figure 7. Dendrogram, Silhouette Score, and Davies–Bouldin Index (DBI) used to determine the optimal number of clusters (K) for hierarchical clustering.

Across both methods, $k = 3$ was the most consistently supported solution — favored by the Silhouette Score and DBI for K-Means and by all three criteria for hierarchical clustering. The Elbow Method alone suggested $k = 4$ as a possible alternative, so we conducted a sensitivity analysis comparing the two partitions under hierarchical clustering.

As shown in Figure 8, the PCA visualizations show that the overall cluster structure remains largely stable between the two solutions. Moving from $k = 3$ to $k = 4$ does not reorganize the clusters; rather, it splits one existing cluster into two. Two clusters ($n = 576$ and $n = 188$) remained unchanged, while the third ($n = 608$) was divided into two smaller groups ($n = 451$ and $n = 157$). The low-mortality cluster ($n = 188$) was fully preserved, and the newly identified subgroup ($n = 157$) corresponds to an extreme high-mortality population. This indicates that $k = 4$ adds resolution within one cluster without changing the broader structure, so we retained $k = 3$ as our primary solution, with $k = 4$ viewed as an optional refinement.

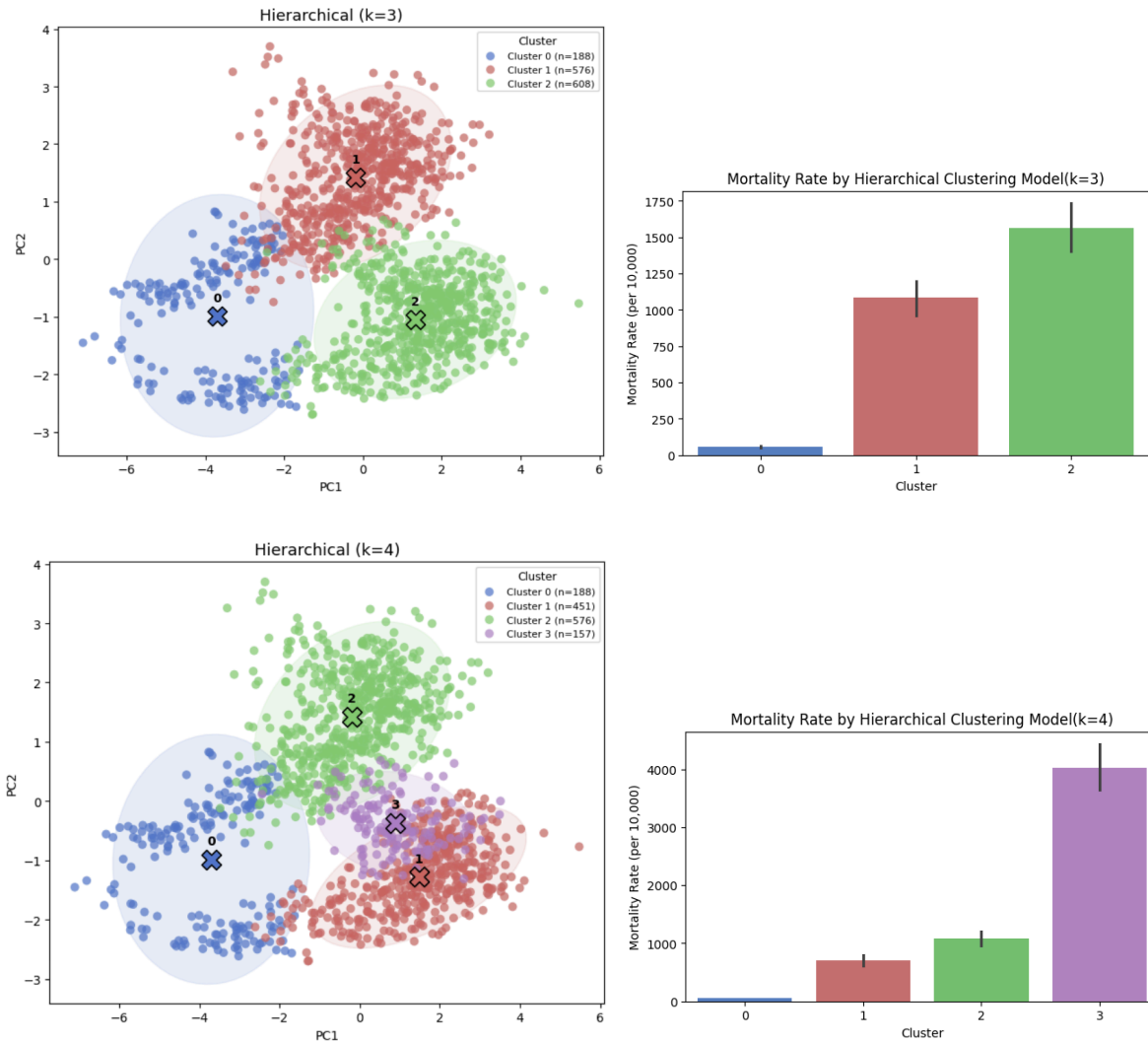


Figure 8. Sensitivity analysis comparing cluster profiles and cluster sizes under $k = 3$ and $k = 4$ solutions.

The three clusters represent distinct population health profiles. Cluster 0 represents a younger, relatively healthy population with the lowest mortality rate. Cluster 1 is made up primarily of older females with moderate cardiometabolic risk and intermediate mortality. Cluster 2 consists of older males with elevated cardiometabolic burden and the highest mortality rate. These differences support $k = 3$ as a reasonable balance between interpretability and parsimony.

Finally, we applied DBSCAN to assess whether the data contains well-defined dense regions and to identify potential outliers. As shown in Figure 9, using the selected tuned parameters, DBSCAN identified two clusters of similar size ($n = 669$ and $n = 672$), separated primarily along the second principal component, along with a small number of noise points ($n = 31$) at the periphery of the feature space.

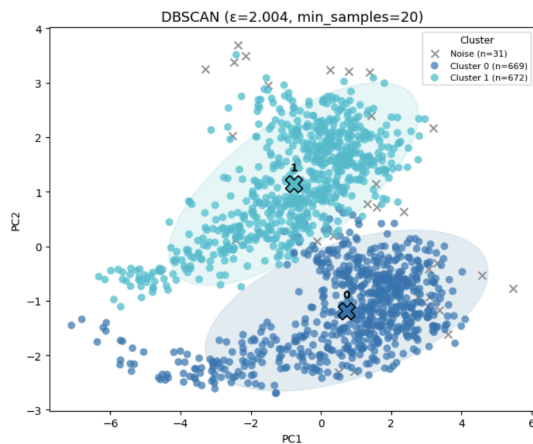


Figure 9. PCA projections of DBSCAN cluster assignments

The low proportion of noise (~2%) suggests the data is largely organized into dense regions rather than scattered outliers. However, the two clusters overlap considerably in the PCA projection and are separated by a diffuse boundary, indicating that the separation between them is moderate rather than sharp. It is also worth noting that DBSCAN's two-cluster solution does not align with the three-cluster solution from K-Means and hierarchical clustering, as DBSCAN divided the data into two broad regions instead.

The variation in results across methods suggests there is no single obvious partition of this dataset. Overall, the findings indicate that population health profiles may vary more along continuous gradients than they fall into sharply distinct subgroups, though some meaningful density structure does exist in the data.

Discussion

Supervised Learning

In the classification tasks, it was unexpected how well RF performed against GBF and SVC. This could be because the depth of hyperparameter tuning for GBF and SVC was limited due to exploding training time. The time to train and fit the more complex models proved to be a time management problem when testing code. To address this, the dataset was reduced to a sample of 1000 rows for much of the testing. Given more time, a survival analysis could have been done to avoid the need to engineer a fixed survival endpoint. This was explored in the classification notebook but was not fully evaluated to the level required for the project. With more time, unsupervised clusters could have been tested as features. Supervised clustering could have been used for feature selection as was shown in Qui et al [2].

In the regression models, one surprising finding was that the Random Forest model produced the largest prediction errors in the oldest age groups (78–90), suggesting that mortality at older ages has greater variability and influenced by additional factors. It could be features like chronic health conditions or broken heart syndrome (one likely to have higher mortality if his/her significant other passes away) that were excluded in our feature engineering. One of the challenges was that Poisson regression had a very high overdispersion, which was

unexpected, since we hypothesized it to be well-suited for mortality prediction. However, due to the overdispersion, we added a Negative Binomial regression model to resolve this issue. If more time was allotted, the project could have incorporated additional health and socioeconomic predictors, performed more extensive sensitivity analysis and hyperparameter tuning, and validated the models more carefully to improve predictive performance and model generalizability.

Unsupervised Learning

One unexpected finding was that different evaluation metrics did not converge on a single optimal value of k . Rather than indicating a flaw in the approach, this highlighted an important limitation of relying on any single metric — even when the Silhouette Score and Davies–Bouldin Index disagree, they can together point to a meaningful range of plausible solutions. Interpreting these conflicting signals was one of the more challenging aspects of this analysis, since the most appropriate choice of k depends partly on the intended use of the clusters, whether that is interpretability, segmentation granularity, or suitability for a downstream task. In response, we shifted from seeking a single "best" k to comparing a small range of solutions and evaluating their stability across metric trends and PCA projections. This revealed that hierarchical structure in the data can support stable macro-clusters while still allowing meaningful subgroups, explaining why both $k = 3$ and $k = 4$ are valid at different levels of resolution.

Given more time, we would explore additional clustering methods — such as Gaussian Mixture Models or HDBSCAN — to better capture non-spherical or density-varying structure. We would also incorporate downstream validation by examining whether clusters differ meaningfully on external outcome variables, ensuring they are substantively interpretable and not only mathematically distinct.

Ethical Considerations

Supervised Learning

Although CDC mortality data and NHANES health data are publicly available and do not contain personal identifying information, there are still important ethical considerations when using them to predict future mortality in supervised learning. The data may reflect existing health inequalities, such as differences in healthcare access or socioeconomic conditions, which could cause the model to produce less accurate predictions for certain socio-economic groups and different ethnicities. In addition, the model is developed using historical data (1999-2017) and may not capture current or future unexpected changes, such as new diseases, future medical advances, or any potential changes in public health policies. Therefore, we recommend that our predictions should only be used for analysis and reference, and not be used as exact forecasts or individual predictions.

Unsupervised Learning

A key issue in unsupervised learning is the misinterpretation of clusters as real or inherent groups. Clustering algorithms will always produce groupings, even when the underlying structure is weak or ambiguous, which can lead to overclaiming that clusters represent "true" categories when they are in fact artifacts of the chosen features, distance metric, and value of k .

To address this, results should be framed as analytical groupings intended for exploration or decision support rather than ground truth. It is also important to communicate uncertainty clearly and demonstrate how results change across different reasonable values of k — as we did through the sensitivity analysis comparing the $k = 3$ and $k = 4$ solutions.

A second concern is bias embedded in the input data and features. If the dataset reflects historical or systemic inequities — whether demographic, socioeconomic, or institutional — clustering may reinforce or amplify those patterns. This is particularly relevant here given that demographic factors such as age and sex directly informed cluster formation, and variables like race/ethnicity and educational attainment were present in the broader dataset. If cluster assignments were used for decision-making such as resource allocation or risk profiling, these groupings could produce unfair or inequitable outcomes. To mitigate this, feature selection should be carefully reviewed, the influence of sensitive attributes on cluster structure should be examined, and results should be evaluated against known inequities before being applied in any consequential context.

Statement of Work

David Ahn	Yinghui Chen	Travis Robinson
• Regression Notebook • Supervised Learning - Regression Section in the final report	• Unsupervised Learning Notebook • Unsupervised Section in the final report • Github Finalization	• Data Cleaning Notebook • Classification Notebook • Introduction, related work, data sources, feature engineering and classification portions of report

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Appendix A. Feature list

Source	Features and Description	
NHANES Full feature descriptions available at https://wwwn.cdc.gov/nchs/nhanes/search/default.aspx	<u>DEMO : Demographics</u>	
	DMDDEDUC2	Education Level – Adults 20+
	RIAGENDR	Gender
	RIDAGEYR	Age at Screening Adjudicated – Recode
	RIDRETH1	Race/Ethnicity – Recode
	SEQN	Respondent sequence number (aggregated to unique count in NHANES+WONDER)
	<u>BMX: Body Measurements</u>	
	BMXWT	Weight (kg)
	BMXHEAD	Head Circumference (cm)
	BMXHT	Standing Height (cm)
	BMXBMI	Body Mass Index (kg/m ²)
	BMXLEG	Upper Leg Length (cm)
	BMXCALF	Maximal Calf Circumference (cm)
	BMXARML	Upper Arm Length (cm)
	BMXARMC	Arm Circumference (cm)
	BMXWAIST	Waist Circumference (cm)
	<u>BPX: Blood Pressure</u>	
	BPXCHR	60 sec HR (30 sec HR × 2)
	BPXSY1	Systolic BP (1st reading) mm Hg
	BPXDI1	Diastolic BP (1st reading) mm Hg

BPXSY2	Systolic BP (2nd reading) mm Hg
BPXDI2	Diastolic BP (2nd reading) mm Hg
BPXSY3	Systolic BP (3rd reading) mm Hg
BPXDI3	Diastolic BP (3rd reading) mm Hg
BPXSY4	Systolic BP (4th reading) mm Hg
BPXDI4	Diastolic BP (4th reading) mm Hg

GLU: Glucose

LBXGLU	Fasting Glucose (mg/dL)
LBDGLUSI	Fasting Glucose (mmol/L)

HDL: Lipids

LBDHDD	Direct HDL-cholesterol (mg/dL)
LBDHDDSI	Direct HDL-cholesterol (mmol/L)

LAB13AM: Lipids

<u>LBXTR</u>	<u>Triglyceride (mg/dL)</u>
<u>LBDTRSI</u>	<u>Triglyceride (mmol/L)</u>
<u>LBDLDL</u>	<u>LDL-cholesterol (mg/dL)</u>
<u>LBDLDLSI</u>	<u>LDL-cholesterol (mmol/L)</u>
<u>SEQN</u>	<u>Respondent Sequence Number</u>

LAB13: Lipids

LBXTC	Total cholesterol (mg/dL)
LBDTC SI	Total cholesterol (mmol/L)
LBDHDL	HDL-cholesterol (mg/dL)
LBDHDLSI	HDL-cholesterol (mmol/L)

LAB10AM: Glucose and Insulin

	LBXGLU	Glucose, plasma (mg/dL)
	LBXIN	Insulin (uU/mL)
	<u>LAB10 and LBXGH: Hemoglobin A1c</u>	
	LBXGH	Glycohemoglobin (%)
	<u>SMQ: Smoking history</u>	
	SMQ020	Smoked at least 100 cigarettes in life
	<u>TRIGLY: Lipids</u>	
	LBXTR	Triglyceride (mg/dL)
	LBDTRSI	Triglyceride (mmol/L)
	LBDLDL	LDL-cholesterol (mg/dL)
	LBDLDLSI	LDL-cholesterol (mmol/L)
	<u>TCHOL: Lipids</u>	
	LBXTC	Total cholesterol (mg/dL)
	LBDTCSI	Total cholesterol (mmol/L)
CDC WONDER Full feature descriptions available at https://wonder.cdc.gov/wonder/help/ucd.html#	year	Year
	age	Age
	gender	Gender
	deaths	Number of deaths
	Crude_rate_per_100k	Crude mortality rate per 100,000 in population
	population	Population count

National Death Index Full feature descriptions available at https://www.cdc.gov/nchs/data/datalinkage/public-use-linked-mortality-files-data-dictionary.pdf		
	SEQN	Respondent Sequence Number
	ELIGSTAT	Eligibility status
	MORTSTAT	Mortality status
	PERMTH_INT	Person-months of follow-up (interval)
	PERMTH_EXM	Person-months of follow-up (exam)
	year	Year